

K均值算法与EM算法

K-means and EM Algorithms

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聚类

Clustering

- ▶ 聚类是一种无监督学习算法，对于给定的一组数据对象的集合，例如 $S = \{x^i | x^i \in \mathcal{R}^n, i = 1, 2, \dots, l\}$ ，我们希望找出数据集 S 中由簇构成的内部结构，满足：

- ▶ 同一簇内的样本彼此相似
- ▶ 与其他簇中的样本不相似

- ▶ 典型应用：

- ▶ 作为独立工具，用于理解和分析数据的分布结构
- ▶ 作为其他算法的预处理步骤

- ▶ Clustering is a unsupervised learning algorithm. For a given dataset $S = \{x^i | x^i \in \mathcal{R}^n, i = 1, 2, \dots, l\}$, We would like to find a structure consisting of clusters within the dataset S , with samples satisfying:

- ▶ Similar to one another within the same cluster
- ▶ Dissimilar to the objects in other clusters

- ▶ Typical applications

- ▶ As a **stand-alone tool** to get insight into data distribution
- ▶ As a **preprocessing step** for other algorithms

K均值算法

K-means Algorithm

- K均值算法尝试将数据划分为 k 个簇，并将数据点分配到各簇，以最小化数据点到其簇中心均值的距离平方和。距离越小意味着相似度越高，因此，同一簇内的数据点彼此相似。该算法通过在两个阶段之间迭代，直至数据点收敛。

K-means algorithm tries to group data into k clusters and attempt to group data points to minimize the sum of squares distance to their central mean. Here smaller distance implies larger similarity, therefore, samples are similar to one another within the same cluster. The algorithm works by iterating between two stages until the data points converge.

- 具体地，对于给定的数据集 $S = \{x^i | x^i \in \mathcal{R}^n, i = 1, 2, \dots, l\}$ 与正整数 k ，记 S 的第 j 个簇的质心为 μ_j ，令 r_{ij} 为一个二值变量，用于指示数据点 x^i 是否属于第 j 个簇。我们的目标是找到一种数据点分组方式与质心向量，使得每个数据点到其对应质心的距离平方和最小。

In detail, for given dataset $S = \{x^i | x^i \in \mathcal{R}^n, i = 1, 2, \dots, l\}$ and a positive integer k , denote the centroid of the j -th cluster as μ_j . Let r_{ij} be a binary variable that indicates the membership of data point x^i is in the cluster j or not. Our goal is to find a grouping of data points and prototype vectors that minimizes the sum of squares distance of each data point.

K均值算法

K-means Algorithm

- 我们需要求解 k 个质心向量 μ_j ($j = 1, 2, \dots, k$)以及每个数据点的簇归属，目标函数为：

Therefore, we would to find the k centroids μ_j and the membership for each data point under the objective function:

$$\min_{r_{ij}, \mu_j} \sum_{i=1}^l \sum_{j=1}^k r_{ij} \|x^i - \mu_j\|_2^2$$

- 由于质心 μ_j 是未知的， r_{ij} 又是二值变量，因此该问题实际上是一个NP完全问题，即时间复杂度为关于 j 的指数函数，因此需要更实际的方法来求解。

Since the centroid μ_j is unknown and r_{ij} is a binary variable, the problem is essentially NP-complete, meaning its time complexity is an exponential function with respect to j . Therefore, a more practical approach is required to solve it.

K均值算法

K-means Algorithm

- 在实际中，我们会将 k 个质心初始化为 k 个不同的随机数据点，然后对每个数据点，按下面规则更新归属变量 r_{ij} :

In practice, we first initialize the k centroids to k distinct random data points. Then for each data point, we update the membership variable r_{ij} :

$$r_{ij} = \begin{cases} 1: j = \underset{k}{\operatorname{argmin}} \|x^i - \mu_k\| \\ 0: j \neq \underset{k}{\operatorname{argmin}} \|x^i - \mu_k\| \end{cases}$$

- 然后根据新的归属变量的值，我们更新新的质心:

Then we update the centroids according to the new membership variables:

$$\mu_j = \frac{\sum_{i=1}^l r_{ij} x^i}{\sum_{i=1}^l r_{ij}}$$

- 我们重复上面过程，直到收敛，即质心不再发生变化。

We cycles between two stages until convergence is reached, aka., centroids become stable.

K均值算法

K-means Algorithm

- ▶ 有时候，不恰当地选取初始质心点，会导致k均值算法收敛时间过长，或收敛到局部最优解。下面方法，可使k均值算法在执行过程中更快，执行结果更优。
 - ▶ 从数据点中随机均匀地选取一个中心点。
 - ▶ 对每个数据点 x^i ，计算 $D(x^i) = \min_{\mu_j \in M} \|x^i - \mu_j\|$ ，即 x^i 与已选的质心点集合M中各个中心点的最小距离。
 - ▶ 采用加权概率分布随机选取一个新数据点作为新中心，点 x 被选中的概率与其 $D(x)$ 的平方成正比。
 - ▶ 重复步骤 2 和 3，直至选出k个中心。
- ▶ Sometimes, inappropriate selection of initial centroids can cause the k-means algorithm to converge too slowly or to a local optimum. The following method can make the k-means algorithm faster during execution and yield better results.
 - ▶ Choose one center uniformly at random from among the data points.
 - ▶ For each data point x^i , compute $D(x^i) = \min_{\mu_j \in M} \|x^i - \mu_j\|$, the distance between x^i and the nearest center in the centroid set M that has already been chosen.
 - ▶ Choose one new data point at random as a new center, using a weighted probability distribution where a point x is chosen with probability proportional to $D(x)^2$.
 - ▶ Repeat Steps 2 and 3 until k centers have been chosen.

K 均值算法

K -means Algorithm

- 下面两幅动图展示了 k 均值算法执行的过程与初始质点的不同选取方法对结果的影响。

The two animations below show the execution process of the k -means algorithm and how different initial centroid selection methods affect the results.



软 K 均值算法

Soft K -means Algorithm

- K 均值算法通常假设每个样本被硬分配到唯一的一个簇，即不允许类别归属存在不确定性，也不允许一个样本属于多个簇。软 K 均值算法放松了这种限制，允许每个样本以特定的概率归属于特定概率分布的类别（归属于所有类别的概率之和必须为 1）。

The two animations below show the execution process of the k-means algorithm and how different initial centroid selection methods affect the results.

- 一般说来， k 均值算法给出的数据集的聚类结果，其结构是离散化的。如果问题本身是离散的，这并没有什么。但有些问题本身是连续的，因此想对样本在样本空间的连续分布情况有一定的了解。而软 K 均值算法结合混合高斯分布衍生出的期望最大化算法，为从连续视角建模样本分布特征提供了途径。

Generally speaking, the clustering results of a dataset by k -means algorithm present a discrete structure, which suits the problem which is discrete by nature. However, some problems are inherently continuous, and we intend to gain an understanding of the continuous distribution of samples in the sample space. The Expectation-Maximization (EM) algorithm, derived from the combination of the soft k -means algorithm and the Gaussian mixture model, provides a way to model the characteristics of sample distributions from a continuous perspective.

混合高斯模型

Gaussian Mixture Model

- 首先说一下什么是混合高斯模型。高斯混合模型是一种概率生成模型，将复杂的连续概率分布表示为 k 个独立的多元（或一元）高斯分布（也称为分量/成分）的加权线性组合。每个分量由自身的均值向量 μ_k 、协方差矩阵 Σ_k 和混合系数 p_k （满足 $1 \leq p_k \leq 1$ 且 $\sum_{k=1}^K p_k = 1$ ）刻画。该模型假设：观测数据点的生成过程为，先以概率 p_k 随机选取一个高斯分量，再从该选中分量对应的高斯分布中抽取样本。

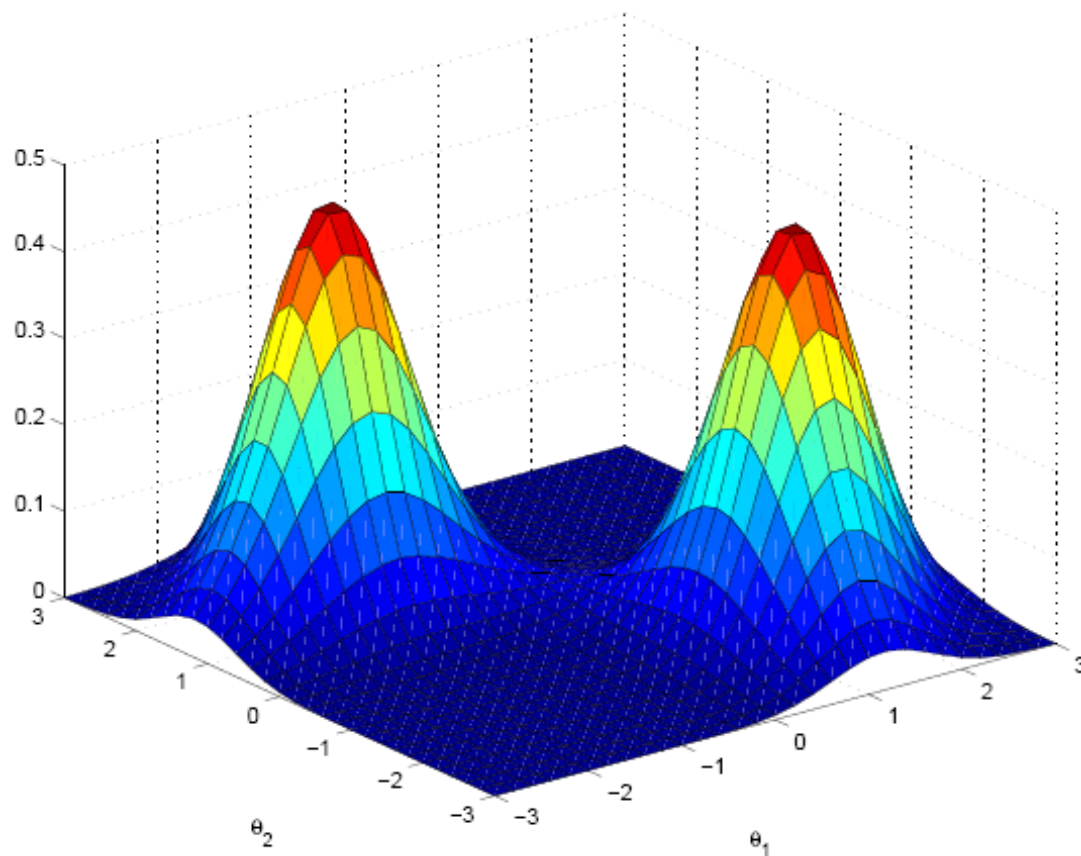
A Gaussian Mixture Model (GMM) is a probabilistic generative model that represents a complex continuous probability distribution as a weighted linear combination of k individual multivariate (or univariate) Gaussian distributions (also called components). Each component is characterized by its own mean vector μ_k , covariance matrix Σ_k , and a mixing coefficient p_k (where $1 \leq p_k \leq 1$ and $\sum_{k=1}^K p_k = 1$). The model assumes that observed data points are generated by first randomly selecting a Gaussian component with probability p_k , then drawing a sample from the selected component's Gaussian distribution.

EM算法

EM Algorithm

- 假设我们的数据集的分布能用混合高斯模型来建模，即 $p(x; \mu, \sigma) = \sum_{k=1}^K p_k \cdot g(x; \mu_k, \sigma_k)$ ，则我们的任务为求解 p_k 、 μ_k 和 σ_k ，使数据点出现的概率达到最大化。

Suppose our dataset can be modeled by GMM, aka., $p(x; \mu, \sigma) = \sum_{k=1}^K p_k \cdot g(x; \mu_k, \sigma_k)$. Our goal then is to find p_k , μ_k , σ_k that maximize the probability of our data points.



EM算法

EM Algorithm

- ▶ EM算法是一种迭代算法，由E步与M步构成：

EM algorithm is an iterative algorithm, consisting of E step and M step.

- ▶ E步（期望步）：计算数据点 x^n 由第 k 个分布生成的概率 $p(k|n)$ ：

E Step: Compute the probability $p(k|n)$ that point x_n is generated by distribution k :

$$p^{(i)}(k|n) = \frac{p_k^{(i)} g(x_n; m_k^{(i)}, \sigma_k^{(i)})}{\sum_{j=1}^K p_j^{(i)} g(x_n; m_j^{(i)}, \sigma_j^{(i)})}$$

- ▶ 在上式中， i 表示第 i 轮迭代；为简化起见，点 x_n 用 n 表示，第 k 个分布用 k 表示。

In the above formula, i denotes the i -th iteration. For simplicity, point x_n is represented by n , and the k -th distribution is represented by k . Then update according to the following formulas:

$$p_k^{(i+1)} = \frac{1}{N} \sum_{n=1}^N p^{(i)}(k|n)$$

EM算法

EM Algorithm

- M步（最大化步）：按最大化似然函数的思路，估计各个分布 k 的参数：

M Step: estimate the parameters for distribution k respectively according to maximum likelihood:

$$m_k^{(i+1)} = \frac{\sum_{n=1}^N p^{(i)}(k|n)x_n}{\sum_{n=1}^N p^{(i)}(k|n)}$$
$$\sigma_k^{(i+1)} = \sqrt{\frac{1}{D} \frac{\sum_{n=1}^N p^{(i)}(k|n) ||x_n - m_k^{(i+1)} ||^2}{\sum_{n=1}^N p^{(i)}(k|n)}}$$

- 在上式中， D 表示样本空间的维数，注意与样本的个数进行区分。

In the above formula, D denotes the dimension of the sample space. Don't mix it with the number of samples.

EM算法

EM Algorithm

- 高斯混合模型的EM算法与K-means算法存在相似之处，其E步对应于簇的分配过程，M步对应于簇中心的更新过程。因此，可以认为K-means是一种简化版的EM算法。在更新模型参数时，K-means做出的是硬决策，而EM算法做出的是软决策。

EM for Gaussian mixtures and K-means shares the similarity, in details, the expectation step is the assignment, and the maximization step is the update of centres. Therefore, K-means can be regarded as a simplified EM, where K-means makes a hard decision while EM makes a soft decision when updating the parameters of the model.

- 注意，与K-均值算法一样，EM算法尽管每一步迭代都会提升在给定分布下数据点的出现概率，但仍可能会陷入局部最优解，虽然这种情况比K-均值算法要少。

Notably, although during each step of EM algorithm, it increases the probability of the points given the distributions. However, the outcome can still get stuck in local optima, though generally it is less than K-Means.